

Hurricane Relief System

Software Design Specification



The Keys

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1. Introduction

Purpose:

The primary purpose of the app should be to allow organization of emergency responses locally. While the average direct death toll for tropical storms is fairly low, at between 10-50 people per year, new studies are showing that hurricanes can be linked to thousands more deaths over the following years, due to resulting homelessness, illness, and sustained injuries.

We want to develop an app that will allow easier management of local resources, as well as connect those in need with those who can help. We also want to make it much easier to allocate resources more accurately and allow the best chance for any individual or community to not only survive, but bounce back.

Source: <https://news.stanford.edu/stories/2024/10/study-links-hurricanes-to-higher-death-rates-long-after-storms-pass>,
<https://www.weather.gov/hazstat/>

2. Stakeholders

- People who live in areas that are at risk of hurricanes
 - Victims
 - Volunteers wanting to help others
- Groups running shelters for hurricane victims
 - Distribution systems for those shelters

Personas



“I’d do anything for my family and community”

Age: 42
Location: Westchase, FL
Additional Info: Father of 2

Scenario

A person requests assistance to get to a shelter. Brad decides to make his way over to them in his truck since they’re already on his family’s way to the same shelter.

Brad Conway

A volunteer to provide assistance

Goals:

- Loves to help people
- Cares about his community

Issues:

- Struggles to find a way to give back

Grew up in Florida, loves the outdoors. Into prepping and survival. Loves his community and cares about his family deeply. Frequently goes camping and hiking. Always stocks extra supplies ‘just in case’. Loves to help others and give back to his community when possible, although he’s unsure how.

Selfless, family/community oriented, skilled outdoorsman

Technology skills

Willingness to try new things

introvert

extrovert



“You should always be prepared when a big storm is approaching”

Age: 30
Location: Tallahassee, Florida
Additional Info: Is a Firefighter

Scenario

He gets called to help people stranded on a roof after helping them get down it turns out that they didn’t know where to go to find shelter as any app they tried wouldn’t show where the nearest shelters were.

Kai Phoenix

A firefighter to help people in trouble

Goals:

- Wants to help spread more awareness for the dangers and help keep people safe.

Issues:

- Feels like people aren’t getting the information and help they need to survive.

Barely lived through a hurricane when younger due to poor information. Was saved by a firefighter and decided to become one after high school and has helped save a lot of people during other storms. He has seen many things that could have been very preventable.

Selfless, Hardworking, Kind Person

Technology skills

Willingness to try new things

introvert

extrovert

	"I love my wife" Age: 26 Location: Miami, Florida Additional Info: Newly married	Scenario John is warned late about a hurricane heading to his area. He is seeking shelter locally and doesn't have much funding to get supplies.
	John Schmungus Potential hurricane victim Goals: - Wants to start a family - Finance a home Issues: - Struggles to get support in his community - Pursuing a job	Forward thinking, classic American ideologies. To-be family man and society-man. Lives in a hurricane/flood-prone area in southern Florida. Attempting to finance a home and family and requires assistance. Wants to help but needs help before he can give it. Puts family first,

Technology skills

Willingness to try new things

introvert extrovert

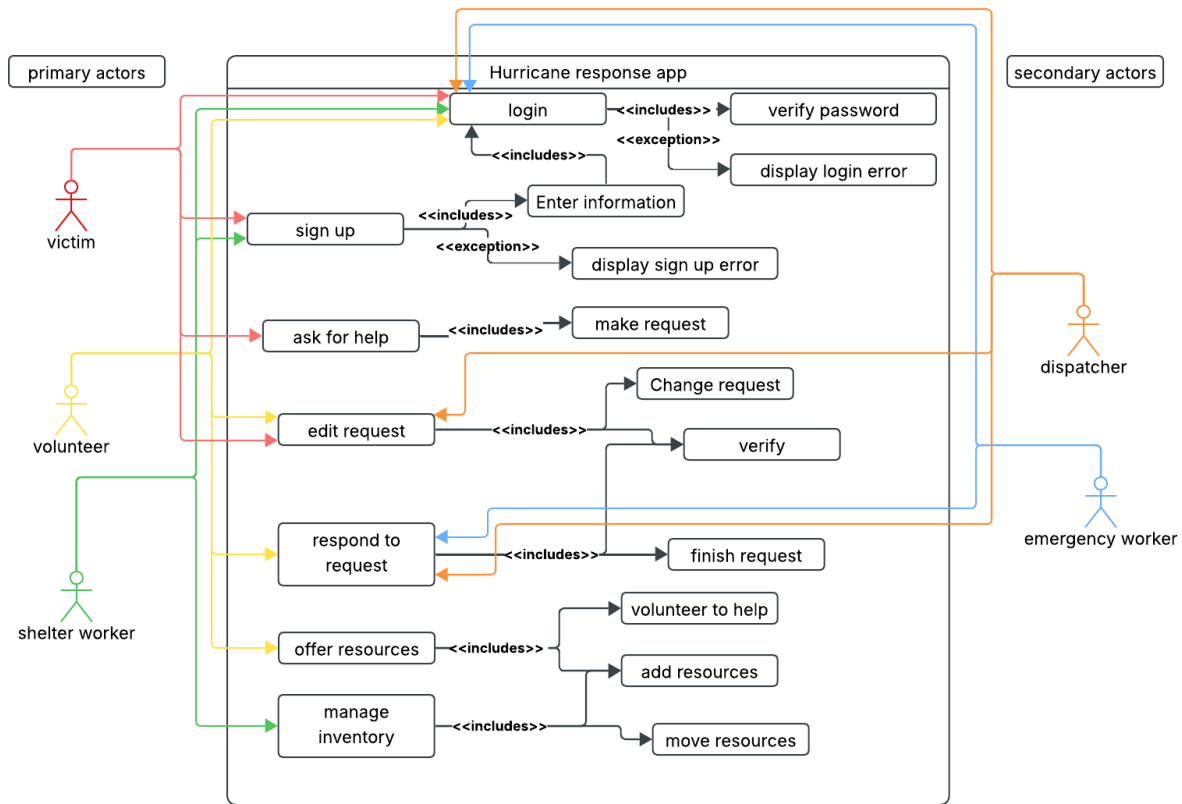
3. Constraints

- Time constraints
 - This project must be finished before the end of the semester.
- Workplace restrictions
 - The group will only regularly meet in person on Tuesdays and Thursdays. Each member will do most of his work on his own time.

4. Overall Description

The app should be accessible in whatever form the users need (mobile or desktop). It should be clearly navigable and efficient to minimize accidental misuse. It needs to be reasonably performant to ensure it doesn't break when a whole local area is using it. The app should display useful info about shelters like capacity and resource stock, display new victims that are in need of assistance, and display who is already being helped.

5. Business Use Cases



6. Functional Requirements

[Requirements Spreadsheet](#)

7. Non-Functional Requirements

- Look and Feel Requirements
 - The system shall be implemented in a way such that its user interface is simple and accessible to users that are not familiar with technology, and is readable to people who might not be able to see very well.
- Performance Requirements
 - The system shall run as long as communication between users and the servers is possible, such as the following:
 - Internet connection
 - Mobile data
- Maintainability and Support Requirements
 - The system shall run on all modern operating systems
 - The system shall run on mobile platforms.
- Security Requirements
 - The system shall keep user data secured
 - The system shall follow best practices for data safety
- Legal Requirements

- The system shall comply with all federal, state, and local laws.

9. Competitive Analysis

	FEMA app
Strengths	The official FEMA app has built in guides to help its users prepare for emergencies. The guides can help people know what to do in preparation for a natural disaster, as well as educate them on what how to act after one happens The app can receive alerts in real time, so people can know very quickly about any weather emergencies that are about to happen. Users can find an emergency shelter directly from the app. This makes it convenient for people who have the app to get somewhere that they can stay. The app is designed to be accessible. It is compatible with screen readers and includes many accessibility features, such as text descriptions. The FEMA app is associated with the government. This makes it the most likely place for people to go to when looking for an app to help with weather and disaster issues.
Weaknesses	Users cannot submit applications for government assistance directly from the app. While the app can help users prepare for applications, they are still expected to submit applications through a separate website. While the app can help users get alerts in real time, the app cannot use the user's gps location for alerts. This means that they must manually pick the locations for which they want alerts for.
Focus	The target for this is people in the US who are likely to be affected by natural disasters. It focuses mainly on preparing people for natural disasters as well as giving information and alerts.

	American Red Cross App
Strengths	Pulls real-time alerts from the National Weather Service for over 40 emergency types into one feed, so the user has a single trusted source instead of checking multiple places. Comes with disaster-focused location tools, including real-time weather maps, live hurricane tracking, and a shelter finder that maps open Red Cross shelters. The map with pins is easy to scan and familiar. Has an "I'm Safe" feature that lets users share their safety status with family in one tap, which adds a personal layer most weather apps lack. Backed by the American Red Cross name and referenced by FEMA and Ready.gov, which increases and verifies its reputation. It is also free, and a lot of the content works offline
Weaknesses	Alert reliability is the biggest complaint. Notifications come late, fail to fire, or trigger for areas that are not relevant, and weak location targeting seems to be the cause. For a safety app, a missed alert defeats the whole purpose. The content is generic and not personalized. It does not adapt to the user's situation, and users looking for real relief like supplies or help will not find it. The recovery side is shallow. There is no real two-way interaction, so you can find a shelter but cannot request aid, report needs, or coordinate with others. This is the main gap our app should target. Smaller issues include battery drain from constant location tracking, crashes during big events, and donation prompts that feel intrusive during an emergency.
Focus	Made for the general public in disaster-prone areas, focused on personal and household preparedness and safety. It centers on the individual getting information, not communities coordinating relief. That is the gap our app can take, moving from one-way alerts toward two-way, community-level relief coordination.

	localrelief.com
Strengths	Has links to a podcast with helpful information and tips, along with links to many news outlets to help out as well. As well as a blog page with even more useful information. Has an easy volunteer form to volunteer to help in whatever way you can by saying your interests in the form. It also allows people to give information on helpful resources or organisations to help. Also has an option to help with money donations in monthly or one time and a partnership program with businesses so they can use promo codes. Has all other expected features of an app like this, like alerts and shelter locations for people to use.
Weaknesses	It is a local non-profit organization that is not backed by any reliable reputation, and it relies on donations to keep running and not a set budget. This could lead to not enough support if something happens. Has had some reliability issues in the past, with people's phone numbers not working when trying to sign up for the app and saying that the phone number doesn't work. It is a smaller weather app and has less support than others and only really covers a local area, so you have to live in that area in order to get the full help with the app.
Focus	Made for fixing a communication breakdown during storms and helping communities find trusted information, local resources, and ways to help people all in one place. It focuses on people helping each other out and keeping people connected before, during, and after storms.

	Strengths	Weaknesses	Focus
FEMA	-Built-in guides -Real-time alerts -Accessibility	-No method of directly requesting assistance. -No GPS tracking for alerts	Victims and people at risk of natural disasters.
Red Cross	-Maps and live tracking -Many location tools -Lets users share their status with family	-Lack of reliability -Not very personalizable -Lack of two-way interaction	Individuals looking for information on disasters.
locarelief.com	-Helpful information links -Multiple ways to volunteer to help -Helpful communication with the community	-Past reliability issues -Smaller weather app and platform -It is a small non-profit not backed by anything	Communication for people to find help for each other.

Overall the competition is fairly strong, although quite fragmented. The goal for this app is mostly to *combine* the strengths of all three, to make a central app that can manage all of this. Someone experiencing a once in a lifetime disaster shouldn't have to juggle 3 or more apps/websites to get help and find out what to do. Our app should ensure that someone can get all the info they need, request assistance, and find out what they need to do next, all in one place. One big (and consistent) weakness is reliability. Our app should be built to be as robust as possible, adding redundancy where possible and making sure if any one piece goes wrong, it doesn't affect the rest of the experience.